DEPARTMENT OF ELECTRICAL ENGINEERING
UNIVERSITY OF ENGINEERING AND TECHNOLOGY PESHAWAR
NOWSHERA CAMPUS

CEP Proposal File

Micro-Processor Based System Design [EE-326]

Group Information

Student Name	Registration No
Muhammad Abuzar Khan	24JZELE**0650**
Mughees ud din	24JZELE**0560**
Khyber Khan	24JZELE**0603**
Muhammad Danish	24JZELE**0570**
Faizan Ullah	24JZELE**0581**

File submitted

To

Dr. Irfan Ahmad

PROJECT PROPOSAL

Smart School Management System

Project Title	Smart School Management System (SSMS)
Project Type	Embedded Systems & Automation
Survey Coverage	7 Schools,Paharpur Region, Pakistan
Date	June 2026
Course	Microcontroller & Embedded Systems

1. Executive Summary

This proposal presents the Smart School Management System (SSMS). It is an embedded-systems project designed to solve key operational problems in Pakistani schools. These problems were identified through a structured field survey conducted across seven schools in the Lar and Rangpur regions.

The system uses microcontrollers to automate four critical school functions. These are: lecture scheduling with an automatic bell, a live teacher-to-class mapping board, an emergency alert system, and a smart student attendance system using a 4x3 keypad for student ID entry. The project addresses real problems faced daily by students, teachers, and school administrations. It is also designed to be expandable in the future with digital applications and advanced features.

2. Background and Motivation

Schools in rural and semi-urban Pakistan face many operational challenges. These challenges reduce the quality of education and create safety risks. Manual processes are slow. Records are lost. Resources are wasted. Communication between administration, teachers, parents, and students is poor.

Technology offers practical solutions to these problems. Microcontrollers, sensors, and wireless communication can automate many school functions. The cost is low. The impact is high. Our team decided to conduct a field survey to understand the real problems. We visited seven schools and collected data. We then selected the most critical and most solvable problems. We designed a system that can be built using the skills and components available in our course.

3. Survey Methodology

3.1 Survey Instrument

A structured questionnaire was designed. It contained 25 questions in both English and Urdu. This made it accessible to all respondents. The questions covered attendance, electricity use, student distraction, emergency preparedness, transport, and technology suggestions.

3.2 Schools Surveyed

The survey was conducted across seven schools. Both government and private schools were included:

#	School Name	Type	Location
1	Government Higher Secondary School Lar	Government	Lar
2	Government Primary School Lar	Government	Lar
3	Sadaqat Public School Lar	Private	Lar
4	Pak Pearl School Rangpur	Private	Rangpur
5	Al-Abbas Science School & College	Private	Bilot
6	Government High School Kacha Mali Kheil	Government	Kacha Mali Kheil
7	Garrison High School	Semi-Government	Kirri khesor

3.3 Respondent Profile

Respondents included both students and teachers. The questions were designed to collect feedback from both groups. This ensured a balanced perspective on school problems.

4. Key Survey Findings

The survey produced clear results. The findings are organized below by topic.

4.1 Academic Challenges

When asked about the most difficult subject, the majority of students gave the following responses:

Rank	Subject	Response Level
1	Mathematics	Highest
2	English	High
3	Science	Moderate

4.2 Operational Problems Identified

Respondents identified many problems. We focused on those that technology can solve. The key problems are:

- Attendance is recorded manually. It takes a lot of time. Records are sometimes lost.
- Electricity is wasted. Fans and lights remain ON in empty classrooms.
- Notebook checking and result formation takes too much time for teachers.
- School records are kept on paper. They are difficult to manage and easy to lose.
- Students are distracted during class by mobile phones, noise, and uncomfortable temperatures.
- School transport has issues with too few seats and lack of student security.

4.3 Suggestions from Respondents

Many respondents gave suggestions for improvement. The most common suggestions were:

1. Smart interactive whiteboards for teachers
2. Homework helper tools or digital resources
3. Automatic attendance system
4. Automatic bell system for class periods

5. Project Description

Based on the survey findings, we have designed the Smart School Management System (SSMS). The system focuses on four core modules. These modules address the most common and most impactful problems. They can all be implemented using microcontrollers, which aligns with our course curriculum.

5.1 Module 1: Automatic Bell System

The automatic bell system replaces the manual ringing of the school bell. A microcontroller stores the full daily schedule. It uses a real-time clock (RTC) module to track time precisely. When a lecture period ends, the system automatically activates the bell. This ensures no time is wasted between periods.

Key features:

- Programmable schedule for different days
- Accurate timing using RTC module
- No human intervention required
- Can support different bell sounds for different events

5.2 Module 2: Live Teacher-Class Mapping Board

A large display board is placed in the school hall. It shows the name of the teacher assigned to each classroom in real time. For example: Class 1 - Sir Ahmed, Class 2 - Ma'am Sara.

Each classroom also has a button. This button is for the class leader. If a teacher does not arrive within 5 minutes of the start of a lecture, the class leader presses the button. The principal's office then receives an alert. An LCD display in the principal's office shows which class is missing a teacher. This improves accountability and reduces wasted lecture time.

Key features:

- Real-time display of teacher assignments
- Classroom button for reporting absent teachers
- LCD alert in the principal's office
- Simple and reliable microcontroller-based design

5.3 Module 3: Emergency Alert System

An emergency button is installed in the school's main hall. When this button is pressed, an emergency siren sounds throughout the school. This provides a fast, clear warning in case of any danger such as fire, medical emergency, or security threat.

Key features:

- Single button activation for speed
- School-wide siren output
- Simple hardware, highly reliable
- Can be extended to multiple zones in the future

5.4 Module 4: Smart Student Attendance System

Each student is assigned a unique numeric ID. When a student arrives, they enter their ID using a 4x3 keypad. The system records their attendance automatically. This eliminates manual attendance completely. It also prevents record loss.

Key features:

- Each student is assigned a unique numeric ID
- Attendance recorded instantly on keypad ID entry
- Data stored in memory for later review
- Eliminates manual errors and lost records
- Fast and suitable for large student populations

6. Reasons for Project Selection

These four modules were selected based on the following criteria:

Criterion	Explanation
Survey Demand	All four modules were directly requested by respondents in the survey.
Feasibility	All modules can be built using microcontrollers covered in our course.
Impact	These problems affect every school, every day. The impact is immediate.
Cost Effectiveness	Microcontroller-based solutions are low-cost and durable.
Scalability	The system is designed to be extended with more features in the future.

Criterion	Explanation
Curriculum Alignment	The project uses skills learned in our Embedded Systems course.

Other problems identified in the survey, such as transport security and interactive whiteboards, were not included. This is because they require hardware and expertise beyond the scope of our current course. We chose to build a focused, functional system that we can fully implement and demonstrate.

7. System Components

The following hardware and software components will be used in the project:

7.1 Microcontroller and Core Electronics

Component	Purpose
8051 Microcontroller / Arduino	Main processing unit for all modules
Real-Time Clock (DS1307/DS3231)	Accurate timekeeping for automatic bell
4x3 Matrix Keypad	Student ID entry for attendance system
Numeric Student ID System	Each student has a unique numeric ID assigned by the system
LCD Display (16x2 or 20x4)	Showing alerts in principal office and mapping board
Push Buttons	Classroom teacher-absent button and emergency button
Buzzer / Siren	Bell output and emergency alarm
LED Indicators	Status indicators for each module
Relay Module	Controlling bell and siren output

7.2 Power and Infrastructure

Component	Purpose
5V DC Power Supply	Powering microcontroller and modules
Connecting Wires and PCB	Circuit connections
Breadboard (for prototyping)	Initial testing and development

7.3 Software Tools

Tool	Purpose
Keil uVision (C51)	Programming 8051 microcontroller in C
Proteus Simulation	Circuit simulation and testing before hardware build
Arduino IDE (if Arduino used)	Programming and uploading code

8. Future Upgradation Possibilities

The project is designed with future expansion in mind. Our teacher emphasized this requirement. The following upgrades are planned for future development:

8.1 Mobile Application for System Control

A mobile application can be developed to allow school authorities to manage the system remotely. Through this app, administrators can change teacher names and room assignments. They can also update the bell schedule without touching the hardware. This makes the system flexible and easy to manage.

8.2 Digital School Record Management

The same application can be extended to manage all school records digitally. This includes student profiles, attendance history, exam results, and teacher records. This directly addresses the survey finding that paper-based record keeping is a major problem.

8.3 Automatic Complaint Management System

A complaint submission system can be added. Students and teachers can submit complaints through a digital interface. The system automatically forwards the complaint to the relevant person. For example, a complaint about a broken fan is forwarded directly to the school electrician. This saves time and ensures complaints are not lost.

8.4 Parent Notification System for Transport

A device installed in the school bus can track students. It can use fingerprint or face recognition technology to detect when a student boards or exits the bus. The system then sends an automatic notification to the student's parents via mobile message. This directly solves the transport security problem identified in the survey.

8.5 Summary of Future Upgrades

Upgrade	Description	Priority
Mobile App	Control system settings remotely via smartphone	Phase 2
Digital Records	Manage all school data in the app	Phase 2
Complaint System	Auto-route complaints to relevant staff	Phase 3

Upgrade	Description	Priority
Bus Tracking & Parent Alerts	Fingerprint/face recognition + parent SMS notifications	Phase 3

9. Expected Outcomes and Benefits

Benefit	Details
Time Saving	Automatic bell and keypad-based attendance eliminate manual tasks.
Improved Accountability	Teacher-absent alerts reduce missed lectures.
Better Safety	Emergency siren provides fast response to crises.
Accurate Records	Keypad-based attendance prevents errors and data loss.
Energy Awareness	Foundation for future smart energy management.
Scalability	System designed to grow with the school's needs.

10. Conclusion

The Smart School Management System is a practical, impactful, and technically sound project. It is built on real data collected from seven schools in Pakistan. It addresses the most commonly reported and most urgent problems.

The four selected modules can be fully implemented using microcontroller technology. They produce immediate benefits for students, teachers, and school administration. The system is also designed to grow into a complete school management platform in the future.

We are confident this project will demonstrate strong technical skills. It will also create real value for schools in our community. We respectfully request approval to proceed with the development phase.

Thank you